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ECHO CLIENT - SERVER USING TCP/UDP SOCKETS

AIM:

To implement an echo client - server using TCP/UDP sockets.

CODE :

```
import socket
import threading
def start_server():
    server = socket.socket(socket.AF_INET,
                           socket.SOCK_STREAM)
    server.bind(('127.0.0.1', 5000))
    server.listen(1)
    Print(f"[SERVER] listening on 127.0.0.1:5000")
    Conn, addr = server.accept()
    while True:
        data = conn.rev(1024).decode()
        if not data (or) data.lower() == 'quit':
            print("[SERVER] connection closed by
                   client.")
            break
        print(f"[SERVER] Received: {data}")
        conn.send(f"server received: {data}".
            encode()
    conn.close()
    server.close()
    def start_client():
        client = socket.socket(socket.AF_INET
        socket.SOCK_STREAM)
```

```
client.send(msg.encode())
if msg.lower() == 'quit':
    break
response = client.recv(1024).decode()
print(f"[SERVER RESPONSE] {response}")
client.close()

if __name__ == "__main__":
    server_thread = threading.Thread(target=start_server, daemon=True)
    server_thread.start()
    start_client()
```

OUTPUT:

```
Enter message (or 'quit' to exit): hello
Received : hello
server received : hello
Enter message (or 'quit' to exit : quit)
connection closed by client.
```

CODE : (UDP)

```
import socket
import threading
def udp_server():
    s = socket.socket(socket.AF_INET, socket_DGRAM)
    s.bind(('127.0.0.1', 12000))
    data, addr = s.recvfrom(1024)
    msg = data.decode()
    if msg.lower() == 'quit'
    print(f"[SERVER] client {addr} ended the connection")
    break
```

```
def upd_client():
    c = socket.socket(socket.AP_INET, socket,
                      sock_DGR_Am)
    server ('127.0.0.1', 12000)
    while True:
        msg = input ("Enter message (or 'quit' to
                      stop):")
        c.sendto (msg.encode(), server)
        if msg.lower () = 'quit':
        break
        data, _ = c.recvfrom (1024)
        print ((([CLINET]) stopped.")
if-name__ = "__main__":
    daemon = True.start(), udp_client().
```

OUTPUT:

```
Enter message (or 'quit' to stop): [SERVER]Running
on 127.0.0.1:12000 hi
Received from ('127.0.0.1', 49670): hi
Echo from server: hi
Enter the message (or 'quit' to stop): hello
Received from : hello
Enter the message (or 'quit to stop): how are you
Echo from server: how are you
Enter message (or 'quit' to stop): quit.
[SERVER] client ('127.0.0.1', 49670) ended the
connection.
```

Result:

Thus, an echo_client server is implemented using TCP & UDP sockets.